

Answer Key & Teacher Notes

Session 1 – Plotting Points on the Coordinate Plane · Session 2 – Reflecting Points Across Axes

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Plotting Points on the Coordinate Plane

TALK PROMPTS

- How do the signs inside an ordered pair tell you which quadrant a point lands in?
- Why does a point with a zero coordinate land on an axis instead of inside a quadrant?

PARTNER TASK

City Map. One partner plots the museum at $(-4, -7)$ and the stadium at $(9, 1)$; the other names each quadrant and describes the moves from the origin. Switch roles with the Park Map: the Pond at $(-8, 4)$ and the Swing Set at $(6, -5)$.

WATCH FOR

- Moving along the y-axis first (plotting $(-5, 4)$ as down 5, right 4).
- Switching x and y, thinking $(-2, 6)$ and $(6, -2)$ are the same point.
- Naming a quadrant for a point on an axis, such as $(0, -5)$.

SESSION 2 • TEACHER NOTES

Reflecting Points Across Axes

TALK PROMPTS

- Why does only one coordinate change sign when a point reflects across an axis?
- How can you name the image before you plot a single point on the grid?

PARTNER TASK

Two Kites. One partner plots Kite A at $(-6, 3)$ and names its reflection across the y-axis; the other finds the left-right distance between the kites (12 units). Switch roles with the Stage Lights: $(-8, 12)$ across the x-axis.

WATCH FOR

- Changing both signs for a single reflection ($(-1, 8)$ written as $(1, -8)$).
- Changing the wrong coordinate (across the y-axis, changing y).
- Swapping the two coordinates instead of changing a sign.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.C.6c	1	2	1	2	2	2	2	3	3	1	1
S1 • Form B	6.NS.C.6c	1	2	1	2	2	2	2	3	3	1	1
S2 • Form A	6.NS.C.6b	1	2	1	1	2	2	2	3	3	1	1
S2 • Form B	6.NS.C.6b	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

coordinate plane: A grid made by a horizontal and a vertical **number line**.

ordered pair: Two numbers (x, y) that name one **point**.

origin: The point **(0, 0)** where the two axes cross.

quadrant: One of the **four** sections of the plane.

- The x-coordinate tells you to move **right or left**. The y-coordinate tells you to move up or down.
- Signs name the quadrant: $(-, +)$ is Quadrant **II** and $(-, -)$ is Quadrant **III**.

Example: Start at the origin, $(0, 0)$. Move 5 units **left**, because $x = -5$. Move 4 units **down**, because $y = -4$. Answer: $(-, -)$ means Quadrant **III**, the lower left.

YOU TRY

1. a. Quadrant IV b. Quadrant II
2. A: 4 left, 2 up (Quadrant II); B: 3 right, 5 down (Quadrant IV)

EXIT TICKET

1. a. Quadrant III b. Quadrant I
2. $(-5, 2)$
3. **B** $((3, -4))$

PRACTICE FORM A

1. a. II b. I c. III d. IV e. on the y-axis f. on the x-axis
2. a. Gate 6 right, 3 up; Pond 4 left, 8 up; Bench 7 left, 2 down; Swing Set 5 right, 6 down b. Gate I, Pond II, Bench III, Swing Set IV c. $(0, 0)$
3. $(-6, 11)$, Quadrant II
4. $(-5, 1)$ is in II; $(1, -5)$ is in IV; not the same point
5. No. $(-2, 6)$ is 2 left and 6 up (Quadrant II). $(6, -2)$ is 6 right and 2 down (Quadrant IV). Order matters.
6. $(-3.5, 2.5)$ in Quadrant II; $(4.5, -1.5)$ in Quadrant IV
7. a. museum III, stadium I b. $(0, 0)$
8. a. school I, park II, pool III b. 3 (school and park) c. 6 units
9. The first number is x, so -5 means 5 units LEFT. The second number is y, so 4 means 4 units UP. The point $(-5, 4)$ is in Quadrant II.
10. **B** (Quadrant II)
11. **B** (right or left)

Self-check word ORIGIN

PRACTICE FORM B

1. a. II b. I c. III d. IV e. on the y-axis f. on the x-axis
2. a. Gate 3 right, 6 up; Pond 8 left, 4 up; Bench 2 left, 7 down; Swing Set 6 right, 5 down b. Gate I, Pond II, Bench III, Swing Set IV c. $(0, 0)$
3. $(-9, -5)$, Quadrant III
4. $(-4, 2)$ is in II; $(2, -4)$ is in IV; not the same point
5. No. $(3, -7)$ is 3 right and 7 down (Quadrant IV). $(-7, 3)$ is 7 left and 3 up (Quadrant II). Order matters.
6. $(-2.5, -4.5)$ in Quadrant III; $(3.5, 1.5)$ in Quadrant I
7. a. museum IV, stadium II b. $(0, 0)$
8. a. school IV, park III, pool II b. -2 (school and park) c. 6 units
9. The first number is x, so -3 means 3 units LEFT. The second number is y, so 6 means 6 units UP. The point $(-3, 6)$ is in Quadrant II.
10. **D** (Quadrant IV)
11. **C** (Quadrant III)

Self-check word POINTS

GUIDED NOTES

reflection: A **flip** that makes a mirror image across a line.

x-axis: The **horizontal** number line through the origin.

y-axis: The **vertical** number line through the origin.

- Across the x-axis, the **y**-coordinate changes sign.
- Across the y-axis, the **x**-coordinate changes sign.

Example: Across the x-axis: keep $x = 7$ and change -2 to 2 . Image: $(7, 2)$. Across the y-axis: keep $y = -2$ and change 7 to -7 . Answer: The images are $(7, 2)$ and $(-7, -2)$.

YOU TRY

1. a. $(3, 5)$ b. $(6, 1)$
2. B = $(-2, -4)$

EXIT TICKET

1. a. $(4, -9)$ b. $(8, -3)$
2. the x-axis; 12 units apart
3. B $((-3, -7))$

PRACTICE FORM A

1. a. $(5, -3)$ b. $(2, 8)$ c. $(7, 1)$ d. $(9, -4)$
2. a. $(4, -7)$, Quadrant IV b. $(-4, 7)$, Quadrant II
3. $(6, 3)$; 12 units
4. $(-8, -12)$; 24 units
5. the x-axis; x stays 2 and only y changes sign
6. He changed y instead of x. Correct image: $(3, 6)$
7. a. $(5, 8)$ b. $(-5, 8)$ c. Quadrant II
8. a. $(-3, 4)$ b. $(3, -4)$ c. $(-3, -4)$: reflect A across both axes (change both signs)
9. A reflection across the x-axis changes only the y-coordinate. x stays -1 and 8 becomes -8 . Correct image: $(-1, -8)$.
10. B $((4, 6))$
11. A (the x-coordinate)

Self-check word REFLECTS

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • PLOTTING POINTS ON THE COORDINATE PLANE

1. A $(4, 3)$ I; B $(-4, 2)$ II; C $(-2, -4)$ III; D $(3, -4)$ IV
2. P $(2, 4)$ in I; Q $(-4, -1)$ in III; S $(-3.5, 3)$ in II; G $(4, -2.5)$ in IV. a. the pool b. the garden
3. a. Quadrant III b. Quadrant IV c. on the y-axis d. Quadrant II e. on the x-axis f. Quadrant IV
4. No. Order matters. $(-3, 5)$ is 3 left and 5 up (Quadrant II). $(5, -3)$ is 5 right and 3 down (Quadrant IV).

PRACTICE FORM B

1. a. $(6, -2)$ b. $(5, 7)$ c. $(8, 3)$ d. $(10, -5)$
2. a. $(-5, -6)$, Quadrant III b. $(5, 6)$, Quadrant I
3. $(7, 2)$; 14 units
4. $(-6, -9)$; 18 units
5. the y-axis; y stays 7 and only x changes sign
6. She changed y instead of x. Correct image: $(-4, -5)$
7. a. $(-6, 3)$ b. $(6, 3)$ c. Quadrant I
8. a. $(-5, 2)$ b. $(5, -2)$ c. $(-5, -2)$: reflect A across both axes (change both signs)
9. A reflection across the x-axis changes only the y-coordinate. x stays -7 and 2 becomes -2 . Correct image: $(-7, -2)$.
10. B $((5, -3))$
11. A $((2, 8))$

Self-check word SYMMETRY

SHEET 2 • REFLECTING POINTS ACROSS AXES

1. a. B = $(-4, -3)$ b. C = $(4, 3)$
2. a. $(6, -2)$ b. $(4, -7)$
3. $(7, 2.5)$
4. K = $(-6, 5)$. Reflecting across the x-axis flips the sign of y, so flip it back: -5 becomes 5 .
5. He changed the y-coordinate. Across the y-axis, y stays the same and x changes sign. Correct point: $(3, -8)$.
6. B $((5, 2))$
7. A (the x-axis)